Recognising materials in Blender renders

Seven classes. Three model families.
Assessment on unseen source groups.

Group members:

The problem and seven classes

Classify material appearance to support browsing of rendered assets



Wood



Metal



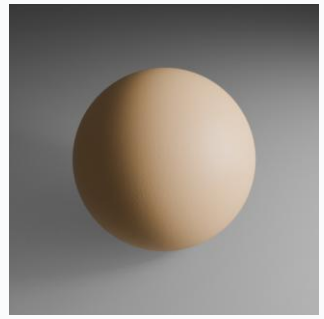
Stone



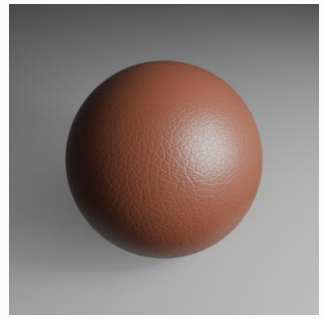
Concrete



Ceramic



Fabric



Leather

One RGB image, one label

Appearance categories do not establish physical composition.

Dataset provenance and group separation

Population	Images	Material IDs	Source groups
Development	3,150	105	101
Fresh test	630	21	21

30 views per material. Each class has 450 development + 90 test images.

Poly Haven + ambientCG CC0 texture assets

Scanned and procedural sources rendered in Blender

Known relatives stay together. Unknown physical ancestry remains a caveat.

Three model families, four pipelines

Pipeline	Transfer learning	Input
Primary fusion	Frozen DINOv2 + ConvNeXt 3,072 features, logistic head	224 + 336
ConvNeXt control	Frozen ConvNeXt-Tiny Logistic head	224
Fixed ResNet-50	ImageNet weights Adapt layer4 and classifier	224
Fine-tuned DINOv2	Pretrained ViT-S/14 + registers Adapt last two blocks and head	336

Architecture, pretraining, resolution and adaptation budgets differ.

Experimental setup

Development and selection

5 outer / 3 inner grouped folds

Frozen features + fold-local scaling

Representation and regularisation search

Seed 20261027

Transfer and preprocessing

RGB, resize/crop, pretrained normalisation

Training only: crops, flips, mild colour jitter

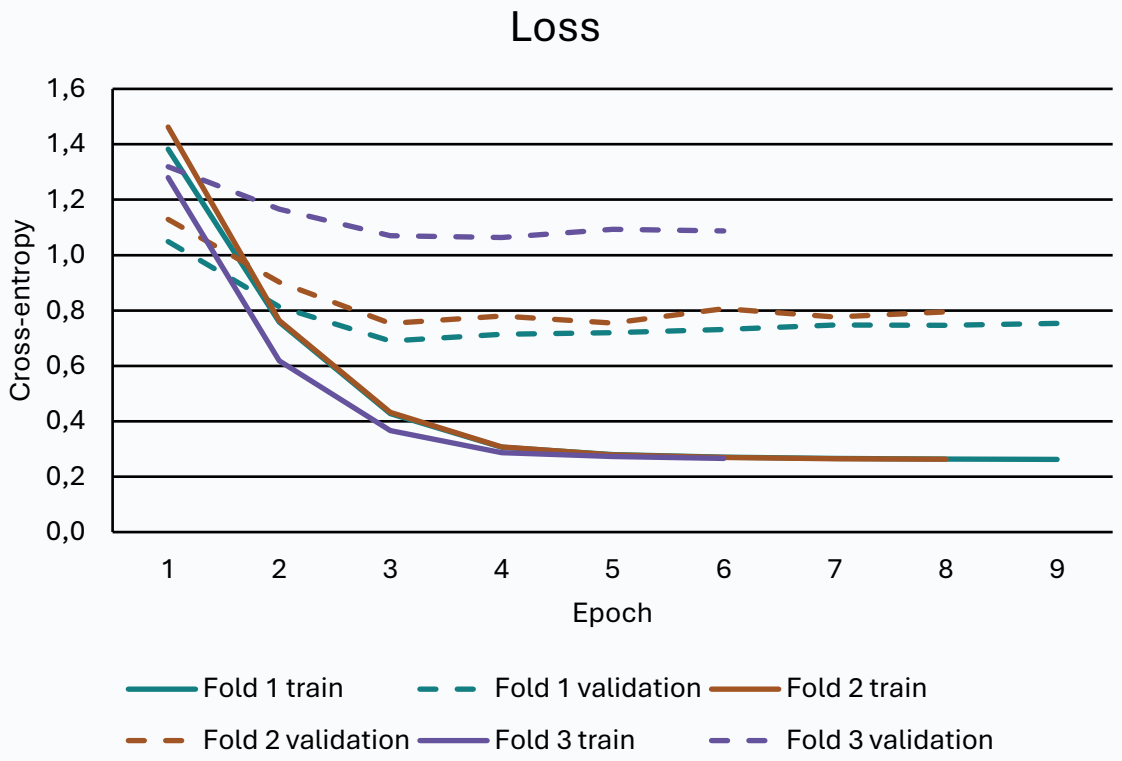
ResNet: 10 fixed epochs

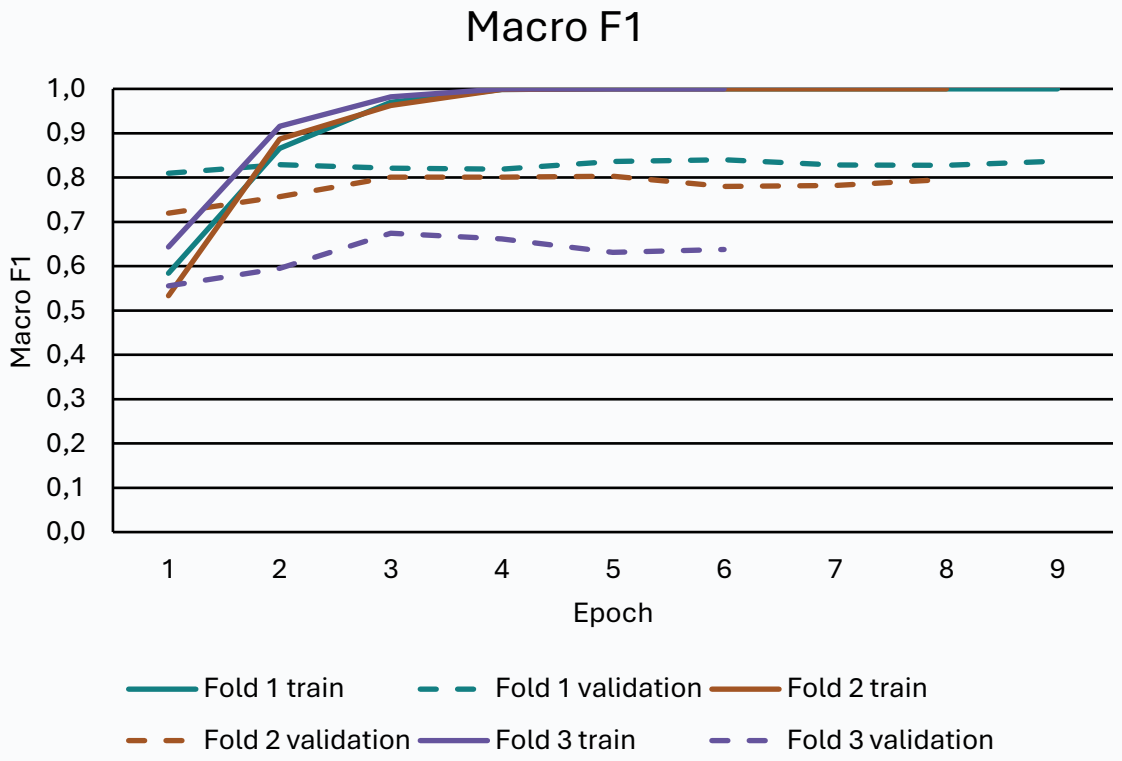
DINOv2: bounded search, 5 final epochs

Choose on development data. Lock every model. Evaluate once on the same 630 images.

DINOv2 adaptation is exploratory. Test ranking does not choose the primary.

Actual learning curves





DINOv2, selected learning rate 1e-5. Solid: training. Dashed: development validation.

Training fit improves faster than validation. Fresh test data never set epochs.

Clean test results on unseen sources

Pipeline	Accuracy	Macro P	Macro R	Macro F1
Primary frozen fusion	80.16%	0.812	0.802	0.7952
ConvNeXt control	75.56%	0.793	0.756	0.7566
Fixed ResNet-50	77.62%	0.789	0.776	0.7665
Fine-tuned DINOv2	86.83%	0.897	0.868	0.8631

Primary: 80.16% accuracy, 95% CI 66.03-93.65%

The 80% point target is met. The lower confidence bound is below 80%.

Exploratory DINOv2 advantage: +6.67 pp, paired 95% CI -2.06 to +16.19 pp.

Higher observed accuracy does not establish superiority.

Primary errors and per-class performance

Confusion counts (90 per true class)

True / pred.	Wd	Mt	St	Co	Ce	Fa	Le
Wood	90	0	0	0	0	0	0
Metal	0	90	0	0	0	0	0
Stone	12	0	60	0	0	0	18
Concrete	18	6	2	61	0	0	3
Ceramic	0	8	30	0	52	0	0
Fabric	0	0	0	0	0	90	0
Leather	0	0	0	22	0	6	62

Wd wood Mt metal St stone Co concrete Ce ceramic Fa fabric Le leather

Precision, recall and F1

Class	P	R	F1
Wood	0.750	1.000	0.857
Metal	0.865	1.000	0.928
Stone	0.652	0.667	0.659
Concrete	0.735	0.678	0.705
Ceramic	1.000	0.578	0.732
Fabric	0.938	1.000	0.968
Leather	0.747	0.689	0.717

Ceramic to stone: 30 errors. Leather to concrete: 22 errors.

Correct and incorrect predictions



True: wood

Predicted: wood

Score: 99.5%



True: metal

Predicted: metal

Score: 96.7%



True: ceramic

Predicted: stone

Score: 96.9%



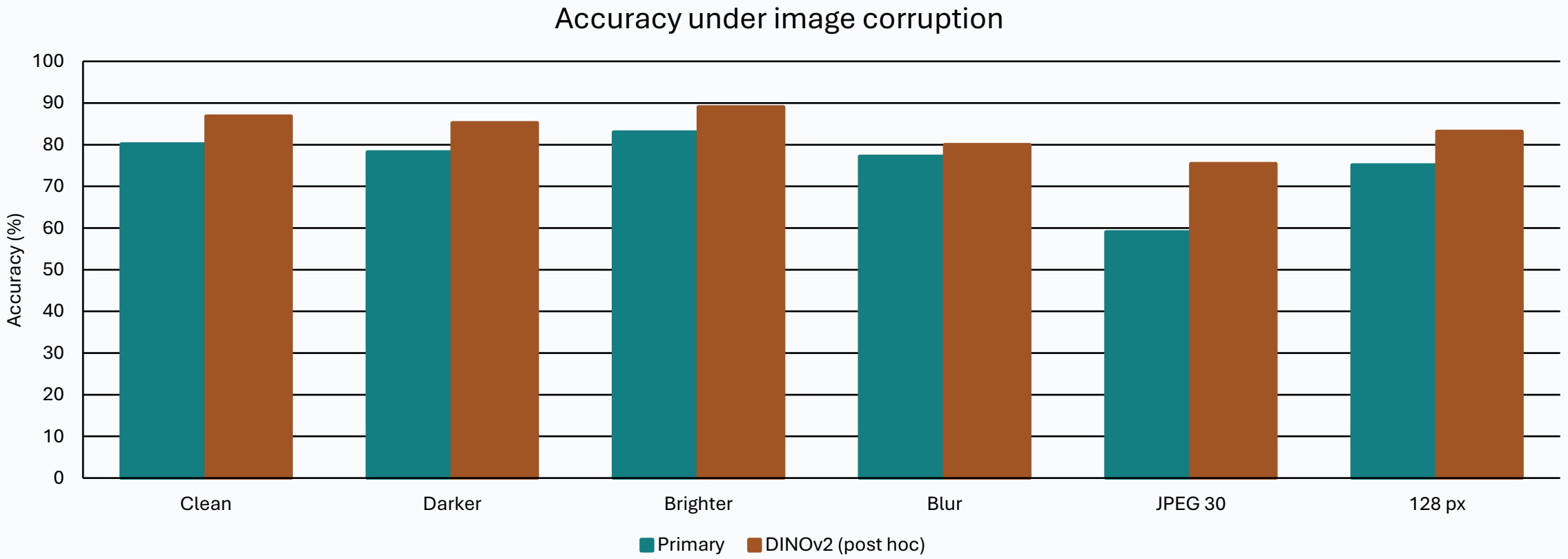
True: leather

Predicted: concrete

Score: 89.9%

A high predicted probability can accompany an incorrect label.

JPEG compression exposes a weakness



Primary: 80.16% clean to 59.05% at JPEG quality 30

DINOv2 corruption results are a post hoc exploratory comparison.

Efficiency and model recommendation

Pipeline	Warm median	Storage
Primary frozen fusion	42.88 ms	203.6 MB
ConvNeXt control	14.20 ms	115.3 MB
Fixed ResNet-50	9.46 ms	95.3 MB
Fine-tuned DINOv2	13.86 ms	88.0 MB

Provisionally recommend fine-tuned DINOv2 for classroom use with comparable renders; retain fusion as the predeclared study primary; independently confirm before broader deployment.

RTX 4070 Laptop GPU. Warm batch 1, predecoded RGB.

Includes preprocessing and probabilities, excludes disk I/O and decoding.

Limitations and next experiment

Few independent sources

Three test groups per class

Related views share texture evidence

Unknown ancestry can remain

Restricted domain

Scanned and procedural assets

Controlled Blender scenes

No real-photograph validation

Next: a fixed evaluation plan with more independent sources, new scenes and photographs, and compression coverage.

Model comparisons do not isolate architecture or equal-compute effects.

Material Studio demo and conclusion

Classify

Image or sample

Choose model

Loading, then probabilities

Results

Stored model comparison

Confusion matrices

Class-level errors

Dataset

Browse the gallery

Inspect source variety

Explain group separation

The primary meets the clean point target.

Wide uncertainty and JPEG weakness limit the claim.

Offline fallback: show stored results, clearly labelled as stored.

Sources

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Liu et al. (2022), A ConvNet for the 2020s
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He et al. (2016), Deep Residual Learning
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Cawley & Talbot (2010), Model selection bias
jmlr.org/papers/v11/cawley10a.html

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Study evidence: saved evaluations, predictions, training histories and source audits.

Exact local sources are cited in slide notes and the speaking guide.